PARTICLE PHYSICS

BY

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RESEARCH AND THEORY
EXACTNESS, DISCRIMINATION, AND THE ACCUMULATION OF MATTER

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For Megan, my loving wife, the only person that really understands what I mean when I speak.

For Autumn and Opal, without you this thought would have never occurred.

To everyone else, I wish you well.

ABSTRACT

The subject: what exact arithmetic lets a reading retain. The method reaches a symbol through equality alone: a point either carries the same symbol as another or it does not, with no order and no distance imposed from the reading end. On integers that equality is exact, and an exact reading never merges two states that differ. This research paper reads the atom that way: an element is the electrons of its neutral atom, each an integer quantum state, and the whole periodic table is one accumulation of them.

Pauli exclusion is an accumulation, and the reading retains it. Two fermions may not share a state. Electrons accumulate into successive states instead of piling into the lowest, and that stacking is the shell structure. The exclusion principle appears here as a count: the same domain-blind primitive that finds two elements on one crystallographic site finds two electrons in one state, and over all 118 elements it finds none. Coarsen the state and distinct electrons agglomerate into one cell; the fullest cell at each coarsening is the Pauli capacity for that level, read off the accumulation with nothing put in.

The periodic law is the recurrence, measured against a drawn null. An element's group signature recurs down the table, and the recurrence is measured by an equality match rate read against two backgrounds: a shuffle that retains the counts and deletes the order, and a filling that retains no exclusion. The recurrence survives neither. It is in the order the exclusion fills the shells and not in the counts, and removing the exclusion leaves a ladder with no table. The row lengths, 8, 8, 18, 18, 32, 32, are read off the shell closures as the differences between them.

What is measured and what is not. The table runs on the ideal filling, Madelung order under Hund's rule, and its figures are deterministic. A later chapter carries the Bohr trajectory in exact rationals, where a radius is n^2/Z and an energy is $-Z^2/n^2$ with nothing rounded, and a whole spectral series reduces to line ratios that carry no measured constant at all. The comparison of the ideal filling with the measured ground states of the NIST database, and of an absolute Bohr line with a measured wavelength, is not reported here. A final chapter takes the Standard Model particles, put in from the particle data, for the structure their exact charges retain. The claims that rest on measurement say so, and the claims that are exact arithmetic say that instead.

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1

EXACTNESS AND INFINITE DISCRIMINATION

Every figure in this chapter is computed on the ideal filling and is deterministic. The ideal filling is Madelung order under Hund's rule, stated in full in the next chapter. What is exact here is the arithmetic, and the figures measure what it retains.

1.1 THE METHOD READS BY EQUALITY

The method [10] reaches a symbol through one operation. Its field interface answers, for two positions in a corpus, whether they carry the same symbol: no order on the symbols, no distance between them, and no alphabet they are drawn from. A measure built on that interface can read an element, a character or a chess move without knowing which, because all it ever asks is whether two things are the same thing.

On integers that question is exact. Two integer quantum states are the same state or they are not, and the reading never has to set a threshold for how close is close enough. That is the property this research paper rests on: where the symbols are exact, the discrimination is unbounded. Nothing the source distinguishes is lost, because nothing is rounded into anything else on the way in.

1.2 A COARSE READING IMPOSES A QUANTUM

The ingestion retains every digit its source writes and imposes no quantum from the reading end. A quantum is a decision about the smallest difference worth telling apart. A crystallographic reading that puts every deposited site on a grid of a quarter of an angstrom makes one silently, and its recovered cell edge carries the grid and not the deposit; the exact reading of the same cells answers the published edge with an error of zero [8]. The difference between the two readings is the grid.

An atom has the same failure available. An electron sits at a quantum state of four numbers, the principal, azimuthal, magnetic and spin numbers, and a reading that retains only the first two, the subshell, drops the two that tell the electrons of one subshell apart. The choice is the voxel again, in a different domain.

1.3 PAULI AS A COUNT

The representation is the electron set of the neutral atom, one point per electron at an integer state. Two electrons at one state is a Pauli violation, and the same domain-blind primitive that

finds two elements on one crystallographic site finds it: the reading that returns the positions carrying more than one value. Over all 118 elements the exact reading places Z electrons at Z distinct states and no state is contested. Exclusion holds for every element the table names.

The check is tested by breaking its subject. A filling that places two electrons at one state retains one entry where the count expects two; the distinct-state count falls below Z and the run reports it. A correct fill of two electrons retains two states; a doubled state retains one. The count separates them, and the labels never disagree, because both electrons carry the same subshell. The loss under a coarse reading is a count and only a count, and a check that compares values reports nothing: across the table the subshell reading collapses 5645 electron states and a value-comparing check flags none of them.

1.4 THE PARTITION IS THE QUANTUM NUMBERS

The partition of an atom is which of the four numbers the reading retains, and the sweep runs over them. Retain all four and every electron is at its own state. Drop the spin and the two electrons of an orbital fall to one cell. Drop the magnetic number and a subshell falls together. Drop the principal number and the atom is one cell.

The fullest one cell ever gets at each partition is the capacity Pauli allows there, and it is read off the accumulation and not put in: one electron per spin orbital, two per orbital, fourteen at the f subshell, thirty two at a full $n = 4$ shell. Only the full partition retains every element's electrons apart. Every coarser one agglomerates, and hydrogen, with one electron to fold, is the only element that survives every partition. Exact arithmetic is the full partition, and the reading that never agglomerates two states that differ is the reading the method is built to be.

2

ACCUMULATION AND THE PERIODIC LAW

The figures here are measured on the ideal filling and are deterministic. The ideal filling is defined first, because every later claim is a claim about it and not about a measured atom.

2.1 THE IDEAL FILLING

The element table builds an element's electrons in Madelung order, lowest principal plus azimuthal first, and Hund's rule inside a subshell, every magnetic orbital taking a spin-up electron before any takes spin-down. The order fills 1s through 7p, and its capacities run to 118 exactly. It names every element from hydrogen to oganesson and no further.

This is the necessary condition, the filling the periodic law would follow if nothing competed with it. Real atoms deviate: chromium, copper and others fill against the order, and their measured configurations [6] are the reference for a comparison this research paper does not report. The representation carries the ideal and says so, so that no reader takes the ideal for a measurement.

2.2 EXCLUSION IS AN ACCUMULATION

Two fermions may not occupy one state. An electron added to an atom therefore cannot join the electrons already in the lowest state; it takes the next state the exclusion leaves open, and the one after that takes the next again. Electrons accumulate into successive states, and the stack of occupied states is the shell structure. Exclusion is not a bookkeeping rule positioned over the atom. It is the reason the atom has structure at all.

That accumulation is faithful here for the reason the first chapter gives. A reading that rounds two close states together loses an electron from the stack and shortens a shell. The exact reading never does, because its discrimination is unbounded, and an accumulation of fermions stays an accumulation and never collapses. Exactness lets the method hold the stack, and exclusion makes the stack tall.

2.3 THE RECURRENCE, AGAINST A DRAWN NULL

The differentiating electron of an element is the last one the order adds, and its group signature is its azimuthal type and its subshell's population with the principal number dropped. Dropping the principal number leaves sodium and lithium reading the same signature, and that recurrence

down the table is the periodic law. It is measured by an equality match rate: how often a signature equals the one a fixed number of elements ahead.

A match rate means nothing without the rate a shuffle of the same signatures reaches. Two backgrounds are drawn, each deleting a property from the data. The first retains the census and deletes the order: a uniform permutation of the same signatures, which fixes every count exactly. Its floor is the real floor, and the match rate at the row lengths falls to it. The recurrence is in the order the shells fill and not in the counts. The second deletes the exclusion: every electron drops to 1s, an element becomes a count of electrons in one state, every signature is distinct, and the match rate and its floor are both zero. Without exclusion there is a ladder of counts and no table. The real accumulation is the positive control, and it is the only arm that carries the recurrence.

2.4 THE ROW LENGTHS, OFF THE DIFFERENCE SET

The recurrence does not sit at one lag. The match rate is above its floor at 32, at 18 and at 8, and at their sums, because the rows of the table are 2, 8, 8, 18, 18, 32, 32 and not one length. A reader reporting a single period is wrong for the majority of the table, the harmonic trap the crystallography and game theory research papers [8, 9] meet from the other side. What is one number per row is the gap between shell closures, read off the signatures: a closure is a filled p subshell, with helium closing the first row.

closure at Z	2	10	18	36	54	86	118
row length		8	8	18	18	32	32

The closures are the noble gases, and the gaps between them are the row lengths, with no row told to the reader. This is the difference-set reading of the crystallography research paper moved to one dimension: a period is a difference between two things that agree, and the differences are the complete candidate set.

3

TRAJECTORIES IN EXACT ARITHMETIC

The reproduction in this chapter is exact and the comparison to measurement is not run. The radii, energies and line ratios below are exact rationals, facts of the Bohr model with no digit approximated. Whether a real atom matches them is a separate comparison, named at the end of the chapter and not carried out.

3.1 THE ORBIT IS A RATIONAL

A hydrogen-like ion is one electron on a nucleus of charge Z. Bohr quantizes the angular momentum at an integer multiple of $\hbar$, and the orbit that follows has radius n^2/Z in units of the Bohr radius and energy $-Z^2/n^2$ in units of the Rydberg. Both are exact rationals, an integer over an integer, built with integer arithmetic on the host as the stand-in for the fixed-width limb integers the device carries.

The quantum number n is an exact integer. Two neighboring orbits differ by

$$\frac{(n+1)^2}{Z} - \frac{n^2}{Z} = \frac{2n+1}{Z},$$

exactly, and no two orbits ever fall on one radius, however large n grows. That is the property a floating point orbit loses: past some n the gap between two orbits falls below the last bit, and the two large orbits merge. The exact reading retains them apart without limit, the same unbounded discrimination the shells are read with, now carrying a trajectory instead of a state.

3.2 ONE HUB, PROPAGATED EXACTLY

Every energy level of every hydrogen-like ion is $E(n, Z) = -\mathrm{Ry}\, Z^2/n^2$, and every radius is $r(n, Z) = a_0\, n^2/Z$. One physical quantity sets each, the Rydberg energy and the Bohr radius, and the whole grid of levels and radii follows from it by an exact rational identity. Compute the hub once and the ladder is fixed to the precision of the hub, with nothing added by the arithmetic that spreads it.

The split this draws is the split a precision program needs. The ratios n^2/Z and $1/n^2$, the integer quantum numbers, the capacities and the closures are mathematical: their uncertainty is exactly zero and there is nothing in them to measure better. The Rydberg energy, the Bohr radius, the fine-structure constant and the electron-to-proton mass ratio are physical: they are measured [5], their uncertainty is irreducible, and they are the only floor the exact quantities are on.

3.3 THE RATIOS CARRY NO CONSTANT

Within one spectral series the hub cancels. A transition from n_2 to n_1 has photon energy proportional to $1/n_1^2 - 1/n_2^2$, and the ratio of two lines of one series is a pure rational with no Rydberg and no meter left in it. The Balmer series, $n_1 = 2$, read as wavelength ratios to its first line:

line	upper n	ratio	decimal
Hα	3	1	1.00000
Hβ	4	20/27	0.74074
Hγ	5	125/189	0.66138
Hδ	6	5/8	0.62500

These ratios are predictions with no measurement in them, exact, and a whole ladder of them is on no physical constant at all. A measured ratio meets them with no constant in the comparison either, and the comparison stays clean.

3.4 WHAT THE COMPARISON NEEDS, AND WHAT THE MODEL IS

The absolute size of an orbit in meters, and any comparison of an absolute line to a measured wavelength, needs the Rydberg constant and the Bohr radius, which are measured and cited instead of derived. That comparison is not run here, because it needs the measured spectra, and no measured wavelength is quoted in this research paper.

The floor on a real spectral line is not the arithmetic, which stays exact, but two things above it. One is the measured constants. The other, larger by far, is the model: the Bohr picture omits fine structure, the Lamb shift and every later correction. The model error dominates the constants' uncertainty long before the arithmetic enters. The exact rationals are the model reproduced without loss. How far the model itself is from a measured atom is the question the comparison exists to answer, and it is left open here.

4

THE PARTICLES, AND THE STRUCTURE THE EXACT NUMBERS RETAIN

The particles here are put in, and the structure is read out. The confirmed content of the Standard Model, six quarks, six leptons, five force carriers and the Higgs boson, comes from the Particle Data Group [7], and nothing in this chapter derives a particle from first principles. The chapter shows how the exact quantum numbers organize themselves once the masses are set aside, and the reading that does it is the same equality the shells are read with.

4.1 CHARGE IN THIRDS, AND NO MASS

Electric charge is quantized in thirds of the electron charge. An up quark is $+2$, a down quark is -1, a charged lepton is -3, a neutrino is 0 and a W is ± 3, all in units of a third, and carried that way each is an exact integer. Spin is doubled in the same manner, a fermion reading 1 and a gauge boson 2, and baryon number is carried in thirds. The particle table holds all of it as integers.

No mass enters. A mass is a measured real number, and reading structure from it would mean choosing a tolerance, a bound on how close two masses must be to count as one. This work refuses a bound everywhere, and it refuses one here: every question the chapter asks is answered by equality or by an exact sum, never by a threshold. Two particles carry the same charge or they do not; a generation's charges sum to zero or they do not.

4.2 THE GENERATIONS RECUR

Strip a fermion's generation and its mass, and its signature remains: its charge, its spin, its color count, its baryon and lepton numbers. The up, charm and top quarks share one signature. The three charged leptons share another. Read over the three generations, the signature sets are equal. The generation is one structure seen three times, the way a periodic group is one signature seen down a column in the second chapter. The only difference between an electron and a muon this reading can see is a mass, and the reading does not look at it.

That recurrence is the periodic table's lesson at a lower level. There the valence signature repeats down a column and the instrument reads the group without being told chemistry. Here the fermion signature repeats across a generation and the instrument reads the family without being told a mass.

4.3 THE CHARGE SUM VANISHES

Sum the electric charge of one generation, counting each quark once for each of its three colors:

$$3 \cdot (+2) \; + \; 3 \cdot (-1) \; + \; (-3) \; + \; 0 \; = \; 0,$$

in thirds, and the same for the second generation and the third. The sum is exactly zero. A vanishing sum is among the conditions for the cancellation of gauge anomalies in the electroweak theory [2]. The three quarks of a proton carry +3 thirds and the electron carries −3, and they balance to the digit; an atom is neutral because of it. That balance is put in here from the particle data. Whether anomaly cancellation alone forces it depends on the fermion content: with three generations and massless neutrinos the charges can be de-quantized, and further structure is needed to fix them [4].

4.4 WHY NO QUARK IS SEEN ALONE

A charge that divides by three is an integer charge. The leptons and the bosons all carry one, and they are the particles seen free. The six quarks do not: their charges are +2 and −1 in thirds, which no count of three divides. A free quark would carry a fractional charge, and none is observed. The combinations that are observed free, the proton at +3 thirds and the neutron at 0 among them, are exactly those whose thirds add to a multiple of three. The arithmetic of the thirds says which combinations carry an integer charge; why only those run free is confinement, which the charge arithmetic does not explain.

4.5 THE HADRONS THE QUARKS MAKE

A baryon is three quarks and a meson is a quark and an antiquark. Over the light quarks u, d and s the ten symmetric three-quark combinations carry the charges of the baryon decuplet, from +2 for the Δ^{++} to −1 for the Ω^{-}, and the nine quark-antiquark combinations carry the pion and kaon charges. Every one of the nineteen carries an integer charge, a multiple of three thirds, and a single quark never does. The observed hadron charges are the exact integer sums of their quark charges, put in from the particle data and added up here, and the proton at +1 balancing the electron is again the neutrality of the atom, read from the quark side. Nothing is derived about why the quarks bind or what a hadron weighs; the charge is a sum of integers.

4.6 THE RAREST IS THE LAST FOUND

The census ranks the quantum numbers by magnitude, the total less a count, the rarest first, the same order the screening probes a field in. Over the eighteen particles the spins fall 1/2 twelve times, 1 five times and 0 once. The scalar is rarest, and the screening reaches the Higgs boson first, the particle observed last [1, 3]. The reading that ranks it first knows nothing of when it was found; it knows only that a spin of zero occurs once.

5

SOURCES AND AVAILABILITY

Each source is listed with what this research paper takes from it.

THE METHOD AND THE READING

The method. The construction this research paper reads with, in its stages of representation, partition, reference, measure, screening and test oracle, and the rule that a symbol is reached through equality alone [10].

Exact ingestion. The primitive that retains every digit and imposes no quantum from the reading end, and the domain-blind readings of which positions are occupied and which carry more than one value, on which the Pauli count is built. Its use on crystal structures is in the crystallography research paper [8].

The element table. One transcribed table of atomic number, symbol and the Madelung filling order, shared with the chemistry research paper.

The particle table. The confirmed Standard Model particles as exact integer quantum numbers, charge in thirds and spin doubled, with no mass.

The equality match rate. The measure the recurrence is read with, and the coincidence floor a shuffle reaches.

REFERENCE DATA

Element symbols and atomic numbers. The IUPAC table, the canonical reference that names the elements 1 to 118. It is standard and carries no single locator; the element table transcribes it and states so.

The Madelung filling order and Hund's rule. Standard atomic structure. The order fills 1s through 7p and its capacities close on 118, which the element table checks and does not assert.

THE MEASURED DATA

NIST Atomic Spectra Database [6]. The measured ground-state configurations of neutral atoms through element 108, which hold the aufbau exceptions the ideal filling does not. The comparison against them is not reported here.

The predicted superheavy configurations. Elements 109 to 118 have no measured neutral atom; their calculated configurations are compiled in [11], marked predicted and retained apart from the measured ones.

The Particle Data Group [7]. The confirmed particle content, three generations and the Higgs boson, and the measured masses, which the particle table records and this reading sets aside.

CODATA recommended values [5]. The Rydberg constant, the Bohr radius and the fine-structure constant, the measured hubs that set the absolute scale the exact rationals are on. The trajectory chapter's absolute comparison needs them and is not run; no value is quoted.

AVAILABILITY

The representation, the measures and the runs that print every figure in this research paper are in the orior repository, `https://github.com/dstroy0/orior`.

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